Age norms and compulsory schooling:

United States, 1850–1900

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Age norms

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Age norms

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When is it too late to change careers?

At what age should I plan to buy a house and start a family?

What skills should children have before leaving primary school?

Age norms and school-entry

Schools institutionalise age norms:

- Curriculum aims to be developmentally appropriate
- Performance-for-age as measure of ability (e.g., standardised testing, 'red-shirting')
- Age a good predictor of academic abilities
- E.g. in Ohio, children enter school between ages 3-6

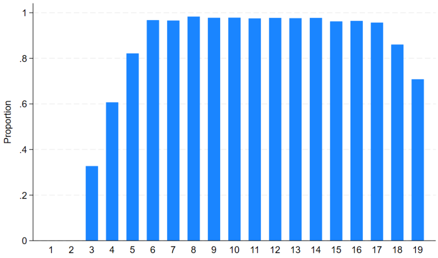

Figure: Proportion of children in school by age, Ohio, 2019

Age norms and school-entry

- Age **not** a good predictor of academic achievement in the past
- In Ohio 1850, children enter school between age 3-9
- Variation in age persisted into higher grades
- Qualitative evidence age norms were very weak in schools historically (Ariès 1964; Vinovskis et al. 1995; Caruso 2023; Lachanski 2025)

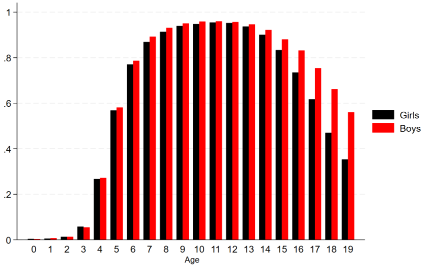

Figure: Proportion of children in school by age and gender, Ohio, 1850

Recitation method (Early 19th century)

Long-established pedagogy. Small groups of children given individualised instruction.

Monitorial/graded instruction (c. 1830-70)

Division of teaching labour. Junior or assistant teachers given elementary subjects. Children classed by ability across subjects.

Age-graded instruction (c. 1900)

Chronological age key feature of classroom structure. Proliferation of age-for-grade statistics after 1900.

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- Lassonde (2013) argues the jump from graded to age-graded is driven by compulsory schooling, which creates a focal entry age.

Theoretical Motivation

- Age grading as part of institutional development of education systems
- Potential downstream effects on human capital (e.g. Cunha & Heckman 2007, Bedard & Dhuey 2011), signalling (e.g. Lenard & Peña 2018), and household labour supply (Manacorda 2006 & Barua 2014)
- The effect of law on norms (Bénabou & Tirole 2025; Leibovitch & Teichman 2025; McAdams 2000; Sunstein 1996)
- E.g. 'normal' pension age and retirement (Seibold 2021; Young 2015)

What is a compulsory schooling law?

- Typical compulsory schooling law defined (Clay et al. 2021):
  - max. school-entry age
  - min. school-leaving age
  - min. period of attendance
  - restrictions on child labour
- Parents free to enroll children before the statutory age

Table 3. Characteristics of initial compulsory education laws

	N (%)
Had a minimum entry-age clause?	
Yes	2 (7.7%)
No	24 (92.3%)
What was the statutory maximum entry age?	
Seven	7 (15.6%)
Eight	23 (71.9%)
Nine	1 (3.1%)
Ten	3 (9.4%)
What was the statutory minimum exit age?	
Thirteen	1 (3.1%)
Fourteen	22 (68.8%)
Fifteen	4 (12.5%)
Sixteen	4 (12.5%)
Eighteen	1 (3.1%)

When were compulsory schooling laws adopted?

- Compulsory schooling laws adopted gradually 1867-1920
- Former confederate states last to adopt, all after 1900

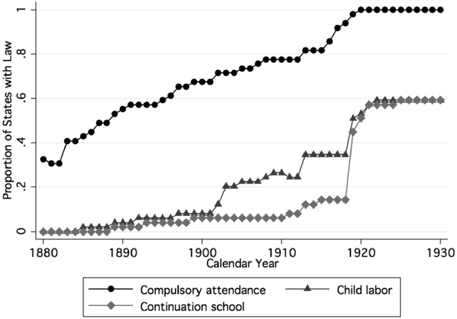


Figure: Cumulative proportion of states with compulsory attendance laws

Source: Clay et al. (2021)

$$E(Y_{i,t(i)}|\{R_{ig}\},\{P_{i,t(i)}\}) = \boxed{\sum_{g=1}^{\overline{g}}\sum_{t=c(g)}^{T}\tau_{gt}R_{ig}P_{i,t(i)}} + \boxed{\sum_{g=1}^{G}\beta_g R_{ig}} + \boxed{\sum_{t=2}^{T}\eta_t R_{i,t(i)}}$$

- Deb et al. (2024) FLEX-DiD estimator for repeated cross-section
- Time and group (state) FEs in not-yet-treated; group- and time-specific TEs averaged to get ATETs
- Allows full flexibility in incorporating covariates, including heterogeneous effects
- For now, heterogeneity by age only
- Assumptions: parallel trends, SUTVA, no bad controls, no anticipation

Data

- Compulsory schooling from Clay et al. (2021) and enrollment from 1% IPUMS (Ruggles et al. 2026):
  - Was child enrolled in school in past year?
- Restrict sample to white, U.S.-born, urban, northern states for comparability with earlier studies and to minimize potential violations of parallel trends
- Years: 1850, 1860, 1870, 1880, & 1900, but **not** 1890.
- Some states have two treatment effects (1880 & 1900), others only one (1900)

Main result

- CS reduces age 4-7 enrollment, where laws did not apply
- Age 10-11 enrollment unchanged
- Implies delayed school entry

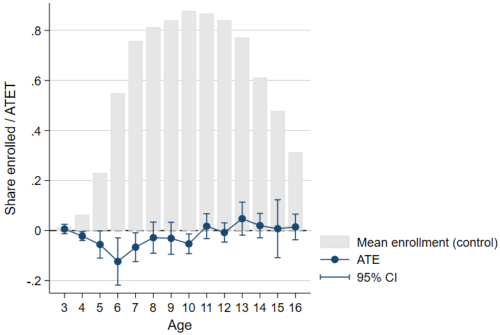
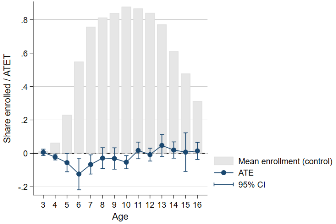


Figure: Effect (ATET) of compulsory schooling on enrollment by age

Summary measure — school-entry age

- Assume enrollment is absorbing state, rising part of age-specific enrollment distribution is CDF
- Mode (a^*) is proportion ever enrolled
- Follows from tail-sum property that:

$$\Delta E(e|\text{enrolled}) = -\frac{1}{F(a^*)} \sum_{a=0}^{a^*} TE(a)$$



Summary measure — school-entry age

Table 2. Effect of compulsory schooling on mean school-entry age by calendar year

	1880	1900
Untreated		
Mean age	6.49	6.86
Proportion ever enrolled	0.84	0.95
N	6,707	2,827
Treated		
Effect on school entry age	0.68 (0.18)	0.31 (0.08)
N	6,229	17,505

- School-entry delayed by 8 months in 1880; 4 months in 1900
- Not-yet-treated mean has increased by 1900, indicating possible behavioural diffusion

Competing explanations — substitution

Table 4. Effect of compulsory schooling on school-leaving age

	1880	1900
Treated		
Effect on school-leaving age	0.01 (0.10)	0.13 (0.11)
N	2,934	8,203

- Law may delay school-leaving, leading to substitution of later for earlier attendance
- Parallel argument to school-entry implies descending part of age-specific enrolment is survival curve for school-leaving
- Compulsory schooling extended school careers by 4 days in 1880 and 47 days in 1900

Competing explanations — state-level policy

SEC. 36. The district schools established under the provisions of this act, shall at all times be equally free and accessible to all children resident therein, over seven and under the age of twenty-one years, subject to such regulations as the district board in each may prescribe. And it shall be the duty of all parents and guardians, or other persons having the control of children between the ages above mentioned, to send such children to some school, at least three months in each and every year, except in case of invalids and others to Sch'ls equally free.

- Qualitative accounts of binding school-entry minima at state level paired with compulsory schooling (Goldin & Katz 2008)
- Re-read all statutes in Clay et al. (2021)
- Identify only two where initial law imposed minimum school-entry age: Wyoming & Montana

Competing explanations — local-level policy

- Possibly local administrators paired age-grading with compulsory schooling laws
- First-grade progression rate as measure of age grading:

$$p = \frac{g_{2,t}}{g_{1,t-1}}$$

- Data only available in some urban districts (Baltimore, Boston, Brooklyn, Cleveland, D.C., New York, Philadelphia, Providence, San Francisco, & St. Louis) and Maryland

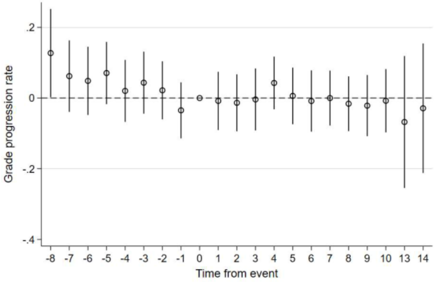


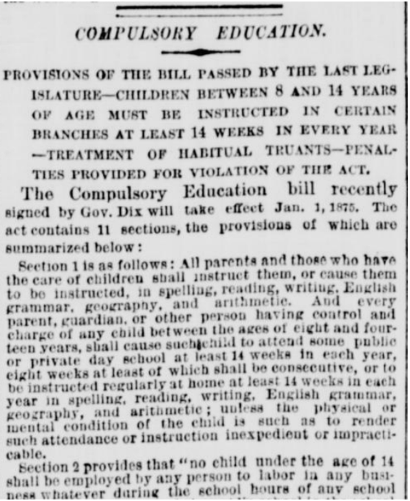
Figure: Response of grade progression to compulsory schooling in select urban districts

Primary salience:

- Individuals will choose salient actions where decision is normatively ambiguous (Mehta et al. 1994; Leibovitch & Teichman 2025)

Secondary salience:

- If coordination is beneficial, individuals may choose actions they believe others will find salient.



COMPULSORY EDUCATION.

PROVISIONS OF THE BILL PASSED BY THE LAST LEGISLATURE—CHILDREN BETWEEN 8 AND 14 YEARS OF AGE MUST BE INSTRUCTED IN CERTAIN BRANCHES AT LEAST 14 WEEKS IN EVERY YEAR—TREATMENT OF HABITUAL TRUANTS—PENALTIES PROVIDED FOR VIOLATION OF THE ACT.

The Compulsory Education bill recently signed by Gov. Dix will take effect Jan. 1, 1875. The act contains 11 sections, the provisions of which are summarized below:

Section 1 is as follows: All parents and those who have the care of children shall instruct them, or cause them to be instructed, in spelling, reading, writing, English grammar, geography, and arithmetic. And every parent, guardian, or other person having control and charge of any child between the ages of eight and fourteen years, shall cause such child to attend some public or private day school at least 14 weeks in each year, eight weeks at least of which shall be consecutive, or to be instructed regularly at home at least 14 weeks in each year in spelling, reading, writing, English grammar, geography, and arithmetic; unless the physical or mental condition of the child is such as to render such attendance or instruction inexpedient or impracticable.

Section 2 provides that "no child under the age of 14 shall be employed by any person to labor in any business whatever during the school hours of any school

New York Tribune, May 16, 1874

Mechanism — secondary salience

"One of the most common errors made by the teacher is to overestimate the intelligence of the over-age pupil. This is because she fails to take account of age differences and estimates intelligence on the basis of the child's school performance in the grade where he happens to be located. She tends to overlook the fact that quality of school work is no index of intelligence unless age is taken into account" (Terman 1916)

- Coordination on school-entry age enables academic achievement to provide better signal of underlying ability

Conclusion

- Raising the school-entry age by 4-8 months an unexpected consequence of compulsory schooling
- The laws did not compel this change, nor did complementary state or local policies
- Leading explanation is that compulsory education provided a focal point (McAdams 2000) for coordination

Thank you!

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Table 1. Year that states first introduced compulsory schooling

Vermont	1867	Oregon	1889
Michigan	1871	Colorado	1889
Connecticut	1872	Utah	1890
Wyoming	1873	New Mexico	1891
Nevada	1873	Pennsylvania	1895
California	1874	Kentucky	1896
Massachusetts	1874	Indiana	1897
Kansas	1874	West Virginia	1897
New York	1874	Iowa	1902
New Jersey	1875	Maryland	1902
Maine	1875	Missouri	1905
Arizona	1875	Delaware	1907
Ohio	1877	Oklahoma	1908
New Hampshire	1879	Tennessee	1913
Wisconsin	1879	North Carolina	1913
D.C.	1880	Texas	1916
Illinois	1883	Louisiana	1916
Rhode Island	1883	Arkansas	1917
Montana	1883	Alabama	1917
South Dakota	1883	Georgia	1917
North Dakota	1883	Virginia	1918
Minnesota	1885	Florida	1919
Washington	1886	South Carolina	1919
Idaho	1887	Mississippi	1920
Nebraska	1887		

Source: Clay et al. (2021)